



MEDIACREST
TRAINING COLLEGE

Advanced Logistic Regression & Classification Theory

Statistical Modeling vs. Machine Learning for Heart Disease
Prediction

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Abstract

This presentation bridges the gap between statistical modeling and machine learning classification, utilizing a cardiovascular dataset to demonstrate robust decision-making frameworks.

- **Foundations:** Clinical data preparation, strategic imputation, and statistical baseline modeling using `statsmodels`.
- **Regularization Theory:** Deep dive into L1 (Lasso) and L2 (Ridge) penalties, including mathematical derivations and geometric interpretations.
- **Machine Learning Transition:** Preprocessing, scaling, and cross-validation strategies using `scikit-learn`.
- **Policy & Interpretability:** Evaluating the trade-off between predictive power (Decision Trees) and clinical transparency (Logistic Regression), culminating in actionable deployment recommendations.

Session Objectives

- Apply clinically sound imputation strategies for missing data (e.g., median for ordinal, explicit categories for nominal).
- Formulate and fit a parsimonious logistic regression model with interaction terms for risk stratification.
- Understand the mathematical and geometric foundations of L1 and L2 regularization.
- Transition from statistical inference (`statsmodels`) to predictive machine learning workflows (`scikit-learn`).
- Evaluate model performance using Stratified K-Fold Cross-Validation and Test Set metrics (ROC-AUC, F1-Score).
- Conduct stratified fairness checks across demographic subgroups.
- Outline governance, validation, and deployment strategies for clinical decision support.

Part 1: Statistical Baseline (statsmodels)

"Establishing inference, interpretability, and clinical assumptions."

Dataset & Dependencies

- **Dataset:** `Heart.csv` (Clinical cardiovascular data, 303 observations).
- **Target Variable:** HD (Heart Disease presence: 0 or 1).
- **Dependencies:**
 - Data Manipulation: `pandas`, `numpy`
 - Statistical Modeling: `statsmodels.api`, `statsmodels.formula.api`
 - Machine Learning: `scikit-learn`
 - Visualization: `matplotlib`, `seaborn`

1.1 Loading the Data

- Clinical datasets often contain inconsistent missing-value representations.
- **Best Practice:** Explicitly define missing value strings during data ingestion to prevent them from being treated as valid categories.

Python Implementation

```
1 import pandas as pd
2
3 # Treat common missing-value strings as missing
4 df = pd.read_csv(
5     'Heart.csv',
6     na_values=['NA', '?', 'NaN', ''])
7 )
8
```

1.2 Handling Missingness: Clinical Reality

Understanding Missing Data Mechanisms

- **MCAR:** Missingness is unrelated to any observed or unobserved data.
- **MAR:** Missingness is related to observed data (e.g., older patients might skip certain tests).
- **MNAR:** Missingness is related to the unobserved value itself (e.g., sicker patients avoid invasive scans).

Clinical Reality

Dropping rows with NA in clinical data can introduce selection bias. We must impute cautiously to avoid losing valuable “missingness” information.

1.2 Imputation Strategy

Variable	Type	Imputation Strategy
Ca	Ordinal (0–3)	Median imputation. Robust to outliers and preserves the ordinal nature of major vessel count.
Thal	Nominal	Create an explicit <code>unknown</code> category. This preserves the information that the test was missing, avoiding bias.

Best Practice

Avoid mean imputation for ordinal clinical scores. Preserving “missingness” as a distinct category in nominal variables prevents the model from making unfounded assumptions.

1.3 Feature Engineering

- **Age Grouping:** Create discrete, clinically meaningful age groups for later stratified analysis.
- **Sex Labels:** Map numerical encodings (0/1) to descriptive string labels for intuitive clinical reporting.

Python Implementation

```
1 import numpy as np
2
3 # Create Age Group
4 df['Age_Group'] = np.where(df['Age'] > 60, '>60', '<=60')
5
6 # Map Sex Labels
7 df['Sex_Label'] = df['Sex'].map({0: 'Female', 1: 'Male'})
8
```

1.4 Target Leakage & Statistical Assumptions

Target Leakage

Ensure no post-diagnosis variables (e.g., results of an angiogram) are used to predict the initial diagnosis. In this dataset, all features are pre-diagnostic clinical measurements.

Statistical Assumptions Checked

- **Linearity of the Logit:** Visually verified using binned Age quartiles.
- **Residual Analysis:** Pearson residuals vs. Fitted values plotted. The characteristic "double-band" structure is expected in binary data. Variance is widest near the center (log-odds ≈ 0), reflecting maximum uncertainty.

Part 2: Transitioning to Machine Learning

"From statistical inference to predictive generalization."

2.1 statsmodels (Inference) vs scikit-learn (Prediction)

statsmodels

- Focus: **Inference** and Interpretability.
- Provides rich statistical diagnostics: p-values, Confidence Intervals, Odds Ratios.
- Lacks built-in ML workflows (scaling, cross-validation, regularization).

scikit-learn

- Focus: **Prediction** and Generalization.
- Optimized for predictive performance on unseen data.
- Requires ALL categories to be represented (no drop_first).
- Handles the intercept internally.

2.2 Preparing Data for ML (Encoding)

- scikit-learn requires all categories to be represented as dummy variables to avoid silent reference-level assumptions.
- We use `drop_first=False` to keep all dummy columns.

Python Implementation

```
1 features_ml = ['Age', 'Sex', 'RestBP', 'Chol', 'MaxHR',  
2               'ExAng', 'Oldpeak', 'Ca', 'ChestPain', 'Thal']  
3  
4 # One-hot encode all categories  
5 X = pd.get_dummies(df[features_ml], drop_first=False, dtype=int)  
6 y = df['HD']  
7
```

2.3 Feature Scaling & Train/Test Split

- **Stratified Split:** Maintains the class ratio (Heart Disease prevalence) in both train and test sets.
- **Feature Scaling:** Crucial for regularization. Distance-based penalties must be applied fairly across variables with different units (e.g., Age vs. Chol).

Python Implementation

```
1 from sklearn.model_selection import train_test_split
2 from sklearn.preprocessing import StandardScaler
3
4 # Stratified split
5 X_train, X_test, y_train, y_test = train_test_split(
6     X, y, test_size=0.25, random_state=42, stratify=y
7 )
8
9 # Scale numeric columns
10 numeric_cols = ['Age', 'RestBP', 'Chol', 'MaxHR', 'Oldpeak', 'Ca']
11 scaler = StandardScaler()
12 X_train_scaled[numeric_cols] = scaler.fit_transform(X_train[numeric_cols])
13 X_test_scaled[numeric_cols] = scaler.transform(X_test[numeric_cols])
14
```

Part 3: Regularization Theory (L1 vs L2)

"Controlling coefficient magnitude and enabling feature selection."

3.1 Why Regularization is Needed

Consider the ordinary least squares (OLS) regression model:

$$y = X\beta + \varepsilon$$

The OLS solution is:

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$$

The Problem

When predictors are highly correlated or when p is large relative to n , OLS coefficients can become:

- Unstable and excessively large.
- Sensitive to small changes in the data.
- Difficult to interpret.

Regularization addresses this by adding a penalty for large coefficients.

3.2 General Regularized Regression Framework

A general penalized regression objective can be written as:

$$\min_{\beta} \left[\underbrace{\|y - X\beta\|_2^2}_{\text{Model fit}} + \lambda \underbrace{P(\beta)}_{\text{Penalty}} \right]$$

where $\lambda \geq 0$ is the regularization parameter.

- $\lambda = 0$: ordinary least squares.
- Small λ : weak regularization.
- Large λ : strong regularization.

The difference between Ridge and Lasso is the choice of $P(\beta)$.

3.3 L2 Regularization — Ridge Regression

Ridge regression uses the squared (L_2)-norm penalty:

$$P(\beta) = \sum_{j=1}^p \beta_j^2$$

Therefore, the objective is:

$$\hat{\beta}_{Ridge} = \arg \min_{\beta} \left[\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right]$$

The fundamental Ridge estimator (ignoring intercept) is derived by setting the derivative to zero:

$$\hat{\beta}_{Ridge} = (X^T X + \lambda I)^{-1} X^T y$$

3.4 Why Ridge Shrinks Coefficients

Suppose predictors are standardized and mutually orthogonal ($X^T X = I$). Then:

$$\hat{\beta}_{Ridge} = (I + \lambda I)^{-1} X^T y = \frac{1}{1 + \lambda} X^T y$$

Since $\hat{\beta}_{OLS} = X^T y$, we obtain:

$$\hat{\beta}_{Ridge,j} = \frac{1}{1 + \lambda} \hat{\beta}_{OLS,j}$$

Important Qualification

Every coefficient is multiplied by the same shrinkage factor in this simplified case. L2 regularization continuously shrinks coefficients toward zero but **generally does not set them exactly equal to zero**.

3.5 L1 Regularization — Lasso Regression

Lasso uses the absolute value of the coefficients (L_1 -norm):

$$P(\beta) = \sum_{j=1}^p |\beta_j|$$

Therefore:

$$\hat{\beta}_{Lasso} = \arg \min_{\beta} \left[\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j| \right]$$

The Crucial Difference

Ridge: $\sum \beta_j^2$ vs Lasso: $\sum |\beta_j|$

3.6 Why L1 Produces Exactly Zero Coefficients

For a single standardized predictor, the Lasso objective simplifies to:

$$L(\beta) = \frac{1}{2}(z - \beta)^2 + \lambda|\beta|$$

Combining the cases for $\beta > 0$ and $\beta < 0$ gives the **soft-thresholding** solution:

$$\hat{\beta}_{Lasso} = S(z, \lambda) = \begin{cases} z - \lambda, & z > \lambda \\ 0, & |z| \leq \lambda \\ z + \lambda, & z < -\lambda \end{cases}$$

Alternatively: $\hat{\beta}_{Lasso} = \text{sign}(z)(|z| - \lambda)_+$ *This equation mathematically explains why Lasso performs feature selection.*

3.7 Geometric Explanation

Regularization can be viewed as constrained optimization:

$$\min_{\beta} \|y - X\beta\|^2 \quad \text{subject to a constraint on } \beta$$

Ridge (L_2)

Constraint: $\sum \beta_j^2 \leq t$

Creates a **circular/spherical** constraint region.

The optimum is smooth and unlikely to land exactly on an axis.

Lasso (L_1)

Constraint: $\sum |\beta_j| \leq t$

Creates a **diamond-shaped** region (in 2D). The key feature is sharp corners located on the coordinate axes. The optimum frequently occurs at a corner, meaning $\beta_j = 0$.

3.8 Derivative Explanation

The mathematical reason can also be seen from the derivatives at $\beta = 0$.

L2 Penalty

$$P(\beta) = \beta^2$$

$$\text{Derivative: } P'(\beta) = 2\beta$$

$$\text{At zero: } P'(0) = 0$$

The penalty's gradient becomes progressively smaller as β approaches zero.

L1 Penalty

$$P(\beta) = |\beta|$$

At zero, the ordinary derivative does not exist.

The subgradient is:

$$\partial|\beta|_{\beta=0} = [-1, 1]$$

This creates a range of values that can balance the loss gradient, making $\beta = 0$ an optimal solution.

3.9 Summary Comparison: Ridge vs Lasso

Property	Ridge (L2)	Lasso (L1)
Penalty term	$\lambda \sum \beta_j^2$	$\lambda \sum \beta_j $
Shrinks coefficients	Yes	Yes
Exact zero coefficients	Generally no	Yes
Feature selection	No	Yes
Handles multicollinearity	Very well	Yes
Geometry	Circle/sphere	Diamond
Typical use	Prediction/stability	Prediction + variable selection

Easy way to remember

Ridge = Shrinks variables (Keep everyone, control influence).

Lasso = Shrinks + removes variables (Keep the strongest, eliminate the weakest).

3.10 Regularization in Practice

Python Implementation

```
1 from sklearn.linear_model import LogisticRegression
2
3 # L2 (Ridge)
4 lr_l2 = LogisticRegression(penalty='l2', C=1.0, solver='lbfgs', max_iter=1000,
5                             random_state=42)
6
7 # L1 (Lasso)
8 lr_l1 = LogisticRegression(penalty='l1', C=1.0, solver='saga', max_iter=1000,
9                             random_state=42)
10
11 lr_l2.fit(X_train_scaled, y_train)
12 lr_l1.fit(X_train_scaled, y_train)
```

Note: L1 regularization successfully performed feature selection by shrinking less important coefficients toward zero, whereas L2 shrank all coefficients proportionally.

Part 4: Comparing Classifiers

"Evaluating generalization and predictive performance."

4.1 Define Models

We compare a linear, regularized model against a non-linear tree-based model.

Python Implementation

```
1 from sklearn.linear_model import LogisticRegression
2 from sklearn.tree import DecisionTreeClassifier
3
4 models = {
5     "Logistic Regression (L2)": LogisticRegression(
6         penalty='l2', C=1.0, solver='lbfgs', max_iter=1000, random_state=42
7     ),
8     "Decision Tree (Non-linear)": DecisionTreeClassifier(
9         max_depth=4, min_samples_split=10, random_state=42
10    )
11 }
12
```

4.2 Cross-Validation Strategy (Stratified K-Fold)

- **Purpose:** Ensures the model generalizes to unseen patients and maintains the class ratio in each fold.
- **Implementation:** 5-Fold Stratified Cross-Validation.

Python Implementation

```
1 from sklearn.model_selection import StratifiedKFold, cross_val_score
2
3 cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
4
5 for name, model in models.items():
6     # Scale data for LR, but NOT for Decision Tree (scale-invariant)
7     X_cv = X_train_scaled if "Logistic" in name else X_train
8     roc_scores = cross_val_score(model, X_cv, y_train, cv=cv, scoring='roc_auc')
9     f1_scores = cross_val_score(model, X_cv, y_train, cv=cv, scoring='f1')
```

4.3 Cross-Validation Results

Model	Mean ROC-AUC	Mean F1-Score
Logistic Regression (L2)	0.895 ($\pm$ 0.054)	0.799 ($\pm$ 0.051)
Decision Tree (Max Depth 4)	0.841 ($\pm$ 0.075)	0.766 ($\pm$ 0.079)

Observation

Logistic Regression provides stable, well-calibrated probabilities. The Decision Tree captures non-linear thresholds but is more prone to variance (higher standard deviation).

4.4 Test Set Evaluation

Model	ROC-AUC	F1-Score	Confusion Matrix
Logistic Regression (L2)	0.934	0.849	$\begin{bmatrix} 34 & 7 \\ 4 & 31 \end{bmatrix}$
Decision Tree (Non-linear)	0.811	0.704	$\begin{bmatrix} 30 & 11 \\ 10 & 25 \end{bmatrix}$

Interpretation

On the independent test set, Regularized Logistic Regression significantly outperforms the Decision Tree in both discrimination (ROC-AUC) and balanced accuracy (F1-Score), while producing fewer False Negatives (4 vs 10).

Part 5: Interpretability vs. Predictive Power

"Balancing clinical transparency with algorithmic performance."

5.1 Explainability Comparison

Logistic Regression

Global, directional, magnitude-based.

Top features by absolute coefficient:

- Sex (1.090)
- ChestPain_asymptomatic (1.046)
- Ca (1.041)
- Thal_normal (-0.858)

Decision Tree

Local, non-linear, hierarchy-based (Gini Importance).

Top features by importance:

- Thal_normal (0.414)
- ChestPain_asymptomatic (0.200)
- Ca (0.132)
- MaxHR (0.094)

5.2 Stratified Analysis: Does performance vary across demographics?

- A model might perform well overall but fail for specific demographic subgroups (algorithmic bias).
- We evaluate the best performing model (Logistic Regression) on different subgroups of the **TEST set**.
- **Subgroups evaluated:** Overall, Male, Female, Age ≤ 60 , Age > 60 .

Fairness Directive

Subgroup-specific discrimination is only one component of model equity. Future validation must also examine sensitivity, specificity, and calibration across these groups before deployment.

5.3 Stratified Fairness Check Results

Subgroup	N	ROC-AUC
Overall	76	0.934
Male	50	0.902
Female	26	0.980
Age $\leq$ 60	58	0.950
Age $>$ 60	18	0.901

Interpretation

The model maintained an ROC-AUC above 0.90 in every reported subgroup. While variation exists (e.g., Female 0.980 vs Male 0.902), the baseline performance remains strong, suggesting no severe immediate algorithmic bias, though continuous monitoring is required.

Part 6: Policy Brief & Recommendations

"From code to clinical impact and governance."

6.1 Executive Summary

- This analysis evaluated whether complex machine learning models provide sufficient predictive gains to justify sacrificing the interpretability of logistic regression for heart disease risk stratification.
- **Finding:** Regularized logistic regression provides the most appropriate balance.
- 5-Fold CV produced a mean ROC-AUC of **0.895** for L2-regularized logistic regression, compared with **0.841** for a decision tree.
- On the independent test set, the selected model achieved an overall ROC-AUC of **0.934**.
- **Policy Recommendation:** Regularized logistic regression should be the primary model for clinical decision support, where transparency, auditability, and clinician trust are essential.

6.2 Key Statistical Findings

The statistical model identified exercise-induced angina (ExAng) and ST depression (Oldpeak) as critical predictors.

Predictor	Odds Ratio	Clinical Interpretation
ExAng	2.80	≈ 2.8 times the odds of heart disease, holding other variables constant.
Oldpeak	2.14	≈ 2.14 times the odds associated with a one-unit increase.

Interaction Effect

An Age × Sex interaction was identified, suggesting that the association between age and heart disease risk is not completely identical across males and females, necessitating sex-specific risk assessment protocols.

6.3 Interpretability-Performance Trade-Off

Model selection should not be based solely on a single predictive metric.

Logistic Regression Offers:

- Transparent coefficients & Odds Ratios.
- Clear direction of associations.
- Straightforward statistical inference.
- Easier auditing and regulatory validation.

Tree-Based/Ensemble ML Offers:

- Ability to model nonlinear relationships.
- Automatic interaction discovery.
- Flexibility in complex prediction problems.

Conclusion: Increased model complexity did not translate into improved performance in this evaluation.

6.4 Recommended Deployment & Validation Strategy

Tiered Deployment Strategy

- 1 **Primary Model:** Regularized Logistic Regression (strong discrimination, transparent, parsimonious).
- 2 **Secondary Models:** Retain complex ML models for benchmarking and potential future ensemble development.

Required Validation Before Operational Use

- **External Validation:** Evaluate on an independent Kenyan clinical dataset.
- **Calibration Assessment:** Ensure predicted probabilities match observed prevalence (Calibration slope/intercept).
- **Subgroup Validation:** Monitor performance by sex, age, and geographic setting.
- **Clinical Utility:** Decision-curve analysis to determine net benefit across threshold probabilities.

Questions?

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